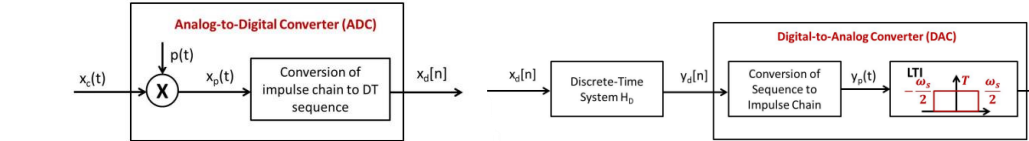


| 性质          | CT Fourier Transform                                                                                  |                                                                                | CT Fourier Series (ω 相同)                                              |                                                        | DT Fourier Transform                                                                                      |                                                                                                              | DT Fourier Series (ω 相同)                                                          |                                                        |
|-------------|-------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------|-----------------------------------------------------------------------|--------------------------------------------------------|-----------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------|--------------------------------------------------------|
| 线性          | $ax(t) + by(t)$                                                                                       | $aX(j\omega) + bY(j\omega)$                                                    | $Ax(t) + By(t)$                                                       | $Aa_k + Bb_k$                                          | $ax[n] + by[n]$                                                                                           | $aX(e^{j\omega}) + bY(e^{j\omega})$                                                                          | $Ax[n] + By[n]$                                                                   | $Aa_k + Bb_k$                                          |
| 时移          | $x(t - t_0)$                                                                                          | $e^{-j\omega t_0}X(j\omega)$                                                   | $x(t - t_0)$                                                          | $a_k e^{-jk\omega_0 t_0}$                              | $x[n - n_0]$                                                                                              | $e^{-j\omega n_0}X(e^{j\omega})$                                                                             | $x[n - n_0]$                                                                      | $a_k e^{-jk\frac{2\pi}{N}n_0}$                         |
| 频移          | $e^{j\omega_0 t}x(t)$                                                                                 | $X(j(\omega - \omega_0))$                                                      | $e^{jM\omega_0 t}x(t)$                                                | $a_{k-M}$                                              | $e^{j\omega_0 n}x[n]$                                                                                     | $X(e^{j(\omega - \omega_0)})$                                                                                | $e^{jM\frac{2\pi}{N}n}x[n]$                                                       | $a_{k-M}$                                              |
| 共轭          | $x^*(t)$                                                                                              | $X^*(-j\omega)$                                                                | $x^*(t)$                                                              | $a_{-k}^*$                                             | $x^*[n]$                                                                                                  | $X^*(e^{-j\omega})$                                                                                          | $x^*[n]$                                                                          | $a_{-k}^*$                                             |
| 时反          | $x(-t)$                                                                                               | $X(-j\omega)$                                                                  | $x(-t)$                                                               | $a_{-k}$                                               | $x[-n]$                                                                                                   | $X(e^{-j\omega})$                                                                                            | $x[-n]$                                                                           | $a_{-k}$                                               |
| 伸缩          | $x(at)$                                                                                               | $\frac{1}{ a }X(j\frac{\omega}{a})$                                            | $x(at)$                                                               | $a_k$<br>$(T \rightarrow \frac{T}{a})$                 | $x_{(k)}[n] = \begin{cases} x[\frac{n}{k}], k \mid n \\ 0, otherwise \end{cases}$                         | $X(e^{jk\omega})$                                                                                            | $x_{(m)}[n] = \begin{cases} x[\frac{n}{m}], m \mid n \\ 0, otherwise \end{cases}$ | $\frac{1}{m}a_k$<br>$(N \rightarrow mN)$               |
| (周期)卷积      | $x(t) * y(t)$                                                                                         | $X(j\omega)Y(j\omega)$                                                         | $\int_T x(\tau)y(t - \tau)d\tau$                                      | $Ta_kb_k$                                              | $x[n] * y[n]$                                                                                             | $X(e^{j\omega})Y(e^{j\omega})$                                                                               | $\sum_{r=(N)} x[r]y[n - r]$                                                       | $Na_kb_k$                                              |
| 乘积          | $x(t)y(t)$                                                                                            | $\frac{1}{2\pi}\int_{-\infty}^{\infty} X(j\theta)Y(j(\omega - \theta))d\theta$ | $x(t)y(t)$                                                            | $\sum_{l=-\infty}^{+\infty} a_lb_{k-l}$                | $x[n]y[n]$                                                                                                | $\frac{1}{2\pi}\int_{2\pi} X(e^{j\theta})Y(e^{j(\omega - \theta)})d\theta$                                   | $x[n]y[n]$                                                                        | $\sum_{l=(N)} a_l b_{k-l}$                             |
| 时域微分 (差分)   | $\frac{d}{dt}x(t)$                                                                                    | $j\omega X(j\omega)$                                                           | $\frac{d}{dt}x(t)$                                                    | $jk\omega_0 a_k$                                       | $x[n] - x[n - 1]$                                                                                         | $(1 - e^{-j\omega})X(e^{j\omega})$                                                                           | $x[n] - x[n - 1]$                                                                 | $(1 - e^{-jk\frac{2\pi}{N}})a_k$                       |
| 时域积分 (累加)   | $\int_{-\infty}^t x(t)dt$                                                                             | $\frac{1}{j\omega}X(j\omega) + \pi X(0)\delta(\omega)$                         | $\int_{-\infty}^t x(t)dt$                                             | $\frac{1}{jk\omega_0}a_k$                              | $\sum_{k=-\infty}^n x[k]$                                                                                 | $\frac{1}{1 - e^{-j\omega}}X(e^{j\omega}) + \pi X(e^{j0})\sum_{k=-\infty}^{+\infty} \delta(\omega - 2\pi k)$ | $\sum_{k=-\infty}^n x[k]$                                                         | $\frac{1}{1 - e^{-jk\frac{2\pi}{N}}}a_k$               |
| 频域微分        | $tx(t)$                                                                                               | $j \cdot \frac{d}{d\omega}X(j\omega)$                                          | —                                                                     | —                                                      | $nx[n]$                                                                                                   | $j \cdot \frac{d}{d\omega}X(e^{j\omega})$                                                                    | —                                                                                 | —                                                      |
| 实信号<br>共轭对称 | $x(t)$ is real                                                                                        | $X(j\omega) = X^*(-j\omega)$<br>...                                            | $x(t)$ is real                                                        | $a_k = a_{-k}^*$                                       | $x[n]$ is real                                                                                            | $X(e^{j\omega}) = X^*(e^{-j\omega})$<br>...                                                                  | $x[n]$ is real                                                                    | $a_k = a_{-k}^*$                                       |
| 实信号<br>奇偶分解 | $\mathcal{E}v\{x(t)\}$<br>$\mathcal{O}d\{x(t)\}$                                                      | $\mathcal{R}e\{X(j\omega)\}$<br>$j \cdot \mathcal{I}m\{X(j\omega)\}$           | $\mathcal{E}v\{x(t)\}$<br>$\mathcal{O}d\{x(t)\}$                      | $\mathcal{R}e\{a_k\}$<br>$j \cdot \mathcal{I}m\{a_k\}$ | $\mathcal{E}v\{x[n]\}$<br>$\mathcal{O}d\{x[n]\}$                                                          | $\mathcal{R}e\{X(e^{j\omega})\}$<br>$j \cdot \mathcal{I}m\{X(e^{j\omega})\}$                                 | $\mathcal{E}v\{x[n]\}$<br>$\mathcal{O}d\{x[n]\}$                                  | $\mathcal{R}e\{a_k\}$<br>$j \cdot \mathcal{I}m\{a_k\}$ |
| 帕斯瓦尔        | $\int_{-\infty}^{\infty}  x(t) ^2 dt = \frac{1}{2\pi} \int_{-\infty}^{\infty}  X(j\omega) ^2 d\omega$ |                                                                                | $\frac{1}{T} \int_T  x(t) ^2 dt = \sum_{k=-\infty}^{+\infty}  a_k ^2$ |                                                        | $\sum_{n=-\infty}^{+\infty}  x[n] ^2 = \frac{1}{2\pi} \int_{-\infty}^{\infty}  X(e^{j\omega}) ^2 d\omega$ |                                                                                                              | $\frac{1}{N} \sum_{n=(N)}  x[n] ^2 = \sum_{n=(N)}  a_k ^2$                        |                                                        |

【连续信号离散化处理】



● Mathematical Interpretation (Fourier Transform)

$x_c(t) \longleftrightarrow X_c(j\omega)$

$x_p(t) = \sum_{n=-\infty}^{+\infty} x_c(nT)\delta(t - nT) \longleftrightarrow X_p(j\omega) = \frac{1}{T} \sum_{k=-\infty}^{+\infty} X_c(j(\omega - k\omega_s))$

$x_d[n] = x_c(nT) \longleftrightarrow X_d(e^{j\omega}) = X_p(j\omega/T)$

Mathematical Interpretation (Fourier Transform)

$y_d[n] = x_d[n] * h_D[n] \longleftrightarrow Y_d(e^{j\omega}) = X_d(e^{j\omega})H_D(e^{j\omega})$

$y_p(t) = \sum_{n=-\infty}^{\infty} y_d[n]\delta(t - nT) \longleftrightarrow Y_p(j\omega) = Y_d(e^{j\omega T})$

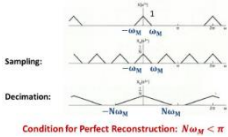
$y(t) = y_p(t) * h_{LP}(t) \longleftrightarrow Y(j\omega) = Y_p(j\omega)H_{LP}(j\omega)$

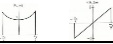
$\tilde{H}_D(e^{j\omega T}) = \begin{cases} H_D(e^{j\omega T}) & |\omega| < \omega_s/2 \\ 0 & otherwise \end{cases}$

It is equivalent to a continuous-time LTI system  $H_C(j\omega) = \tilde{H}_D(e^{j\omega T})$   
 $H_D(e^{j\omega T})$  is a periodic extension of  $\tilde{H}_D(e^{j\omega T})$  with period  $\omega_s = 2\pi/T$

$$\begin{aligned} Y(j\omega) &= Y_p(j\omega)H_{LP}(j\omega) = Y_d(e^{j\omega T})H_{LP}(j\omega) \\ &= X_d(e^{j\omega T})H_D(e^{j\omega T})H_{LP}(j\omega) \\ &= X_p(j\omega)H_D(e^{j\omega T})H_{LP}(j\omega) \\ &= \left[ \frac{1}{T} \sum_{k=-\infty}^{+\infty} X_c(j(\omega - k\omega_s)) \right] H_D(e^{j\omega T})H_{LP}(j\omega) \\ &= X_c(j\omega)H_D(e^{j\omega T}) \\ &= X_c(j\omega)\tilde{H}_D(e^{j\omega T}) \end{aligned}$$

$\tilde{H}_D(e^{j\omega T}) = \begin{cases} H_D(e^{j\omega T}) & |\omega| < \omega_s/2 \\ 0 & otherwise \end{cases}$



| 插值     |                                                                                               |                                                                                       |
|--------|-----------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------|
| 类型     | 插值函数                                                                                          | 还原到理想插值                                                                               |
| 零阶保持   |          |  |
| 一阶保持   |  (频域谱见下) | 总之就是 $\frac{H_{LP}}{H_1}$                                                             |
| 理想 LPF |          | —                                                                                     |

| 系统性质                        |                                             |
|-----------------------------|---------------------------------------------|
| Memoryless 只关于现在            | $h(t) = K\delta(t)$                         |
| Invertible 可逆               | 若有一对多则不可逆                                   |
| Causal 因果                   | $h(t) = 0, t < 0$                           |
| Stable<br>输入有界则输出有界         | $\int_{-\infty}^{\infty}  h(t) dt < \infty$ |
| Time-invariant<br>输入时移则输出时移 | 定义判断                                        |
| Linearity 线性                | 定义判断                                        |

| 连续傅里叶级数收敛条件                            |                                     |
|----------------------------------------|-------------------------------------|
| $\int_T  x(t) ^2 dt < \infty$ (周期能量有界) |                                     |
| 或<br>迪利克雷<br>条件                        | $\int_T  x(t) dt < \infty$ (周期面积有限) |
|                                        | 一定范围内最大最小值个数有限<br>(没有无穷震荡出现)        |
|                                        | 一定范围内不连续点个数有限                       |

| 某低通滤波器                                                                                                                           |
|----------------------------------------------------------------------------------------------------------------------------------|
| $y[n] = \frac{1}{N + M + 1} \sum_{k=-N}^M x[n - k]$                                                                              |
| $H(e^{j\omega}) = \frac{1}{N + M + 1} e^{j\omega \frac{N-M}{2}} \frac{\sin[\frac{\omega}{2}(M + N + 1)]}{\sin \frac{\omega}{2}}$ |

【采样相关】  
Sampling function:  $p(t) = \sum_{n=-\infty}^{\infty} \delta(t - nT)$   
Sampling period:  $T$   
Sampling:  $\omega_s = \frac{2\pi}{T} > 2\omega_M$

Aliasing:  $\cos(\omega_s - \omega_0)t$

$x_p(t) = x(t) \times p(t) = x(t) \times \sum_{n=-\infty}^{\infty} \delta(t - nT) = \sum_{n=-\infty}^{\infty} x(nT)\delta(t - nT)$

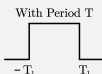
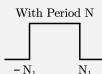
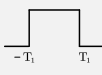
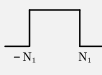
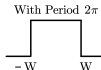
$P(j\omega) = 2\pi a_k \sum_{k=-\infty}^{+\infty} \delta(\omega - k\omega_s) = \frac{2\pi}{T} \sum_{k=-\infty}^{+\infty} \delta(\omega - k\omega_s)$

$P(e^{j\omega}) = \frac{2\pi}{N} \sum_{k=-\infty}^{\infty} \delta(\omega - k\omega_s) \text{ where } \omega_s = 2\pi/N$

$X_p(e^{j\omega}) = \frac{1}{2\pi} \int_{2\pi} P(e^{j\theta})X(e^{j(\omega - \theta)})d\theta = \frac{1}{N} \sum_{k=0}^{N-1} X(e^{j(\omega - k\omega_s)})$

$X_0(e^{j\omega}) = \sum_{n=-\infty}^{\infty} x_0[n]e^{-j\omega n} = \sum_{n=-\infty}^{\infty} x_p[n]e^{-j\omega n/N} = X_p(e^{j\omega/N})$

$= \frac{1}{N} \sum_{k=0}^{N-1} X(e^{j(\omega/N - k\omega_s)})$

| CT Signal                                                                                                                                                           | DT Signal                                                                         | CT Fourier Transform                                                                   | CT Fourier Series                                    | DT Fourier Transform                                                                                                   | DT Fourier Series                                                                                                                                     |                       |
|---------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------|----------------------------------------------------------------------------------------|------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------|
| $e^{j\omega_0 t}$                                                                                                                                                   | $e^{j\omega_0 n}$                                                                 | $2\pi\delta(\omega - \omega_0)$                                                        | $a_1 = 1$<br><i>otherwise</i> 0                      | $2\pi \sum_{l=-\infty}^{+\infty} \delta(\omega - \omega_0 - 2\pi l)$                                                   | $a_k = 1, k = m, m \pm N, \cdots$<br><i>otherwise</i> 0                                                                                               | 周<br>期<br>信<br>号      |
| $\cos \omega_0 t$                                                                                                                                                   | $\cos \omega_0 n$                                                                 | $\pi[\delta(\omega - \omega_0) + \delta(\omega + \omega_0)]$                           | $a_1 = a_{-1} = \frac{1}{2}$<br><i>otherwise</i> 0   | $\pi \sum_{l=-\infty}^{+\infty} \{\delta(\omega - \omega_0 - 2\pi l) + \delta(\omega + \omega_0 - 2\pi l)\}$           | $a_k = \frac{1}{2}, k = \pm m, \pm m \pm N, \cdots$<br><i>otherwise</i> 0                                                                             |                       |
| $\sin \omega_0 t$                                                                                                                                                   | $\sin \omega_0 n$                                                                 | $\frac{\pi}{j}[\delta(\omega - \omega_0) - \delta(\omega + \omega_0)]$                 | $a_1 = -a_{-1} = \frac{1}{2j}$<br><i>otherwise</i> 0 | $\frac{\pi}{j} \sum_{l=-\infty}^{+\infty} \{\delta(\omega - \omega_0 - 2\pi l) - \delta(\omega + \omega_0 - 2\pi l)\}$ | $a_k = \frac{1}{2j}, k = m, m \pm N, \cdots$<br>$a_k = -\frac{1}{2j}, k = -m, -m \pm N, \cdots$<br><i>otherwise</i> 0                                 |                       |
| 1                                                                                                                                                                   |                                                                                   | $2\pi\delta(\omega)$                                                                   | $a_0 = 1$<br><i>otherwise</i> 0                      | $2\pi \sum_{l=-\infty}^{+\infty} \delta(\omega - 2\pi l)$                                                              | $a_k = 1, k = 0, \pm N, \cdots$<br><i>otherwise</i> 0                                                                                                 |                       |
|                                                                                     |  | $\sum_{k=-\infty}^{+\infty} \frac{2 \sin k\omega_0 T_1}{k} \delta(\omega - k\omega_0)$ | $\frac{\sin k\omega_0 T_1}{k\pi}$                    | $2\pi \sum_{l=-\infty}^{+\infty} a_k \delta(\omega - \frac{2\pi k}{N})$                                                | $a_k = \frac{\sin \left[ \frac{k\pi}{N} (2N_1 + 1) \right]}{N \sin \frac{k\pi}{N}}, k \neq 0, \pm N, \cdots$<br><i>otherwise</i> $\frac{2N_1 + 1}{N}$ | 非<br>周<br>期<br>信<br>号 |
| $\sum_{n=-\infty}^{+\infty} \delta(t - nT)$                                                                                                                         | $\sum_{n=-\infty}^{+\infty} \delta[t - kN]$                                       | $\frac{2\pi}{T} \sum_{k=-\infty}^{+\infty} \delta(\omega - k\omega_0)$                 | $a_k = \frac{1}{T}$                                  | $\frac{2\pi}{N} \sum_{k=-\infty}^{+\infty} \delta(\omega - \frac{2\pi k}{N})$                                          | $a_k = \frac{1}{T}$                                                                                                                                   |                       |
|                                                                                     |  | $\frac{2 \sin \omega T_1}{\omega}$                                                     | —                                                    | $\frac{\sin \left[ \frac{\omega}{2} (2N_1 + 1) \right]}{N \sin \frac{\omega}{2}}$                                      | —                                                                                                                                                     |                       |
| $\frac{\sin Wt}{\pi t}$                                                                                                                                             | $\frac{\sin Wn}{\pi n}$                                                           |       | —                                                    |                                       | —                                                                                                                                                     |                       |
| $\delta(t)$                                                                                                                                                         | $\delta[n]$                                                                       | 1                                                                                      | —                                                    | 1                                                                                                                      | —                                                                                                                                                     |                       |
| $u(t)$                                                                                                                                                              | $u[n]$                                                                            | $\frac{1}{j\omega} + \pi\delta(\omega)$                                                | —                                                    | $\frac{1}{1 - e^{-j\omega}} + \pi \sum_{k=-\infty}^{+\infty} \delta(\omega - 2\pi k)$                                  | —                                                                                                                                                     |                       |
| $e^{-at}u(t)$                                                                                                                                                       | $a^n u[n]$                                                                        | $\frac{1}{a + j\omega}$                                                                | —                                                    | $\frac{1}{1 - ae^{-j\omega}}$                                                                                          | —                                                                                                                                                     |                       |
| $\frac{t^{n-1}}{(n-1)!} e^{-at}u(t)$                                                                                                                                | $\frac{(n+r-1)!}{n! (r-1)!} a^n u[n]$                                             | $\frac{1}{(a + j\omega)^n}$                                                            | —                                                    | $\frac{1}{(1 - ae^{-j\omega})^r}$                                                                                      | —                                                                                                                                                     |                       |
| P.S. 以上最后两条 CT 下 $\Re\{a\} > 0$ 或 DT 下 $ a  < 1$ ; DTFS 默认满足周期条件即 $\frac{\omega_0}{2\pi}$ 有理; CT 下 $\omega_0 = \frac{2\pi}{T}$ 或 DT 下 $\omega_0 = \frac{2\pi m}{N}$ |                                                                                   |                                                                                        |                                                      |                                                                                                                        |                                                                                                                                                       |                       |

| 定义式及常用公式 |                                                                                                                                                                                                                                                                                                   |                                                                   |
|----------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------|
| CTFS     | $x(t) = \sum_{k=-\infty}^{+\infty} a_k e^{jk\omega_0 t}$                                                                                                                                                                                                                                          | $a_k = \frac{1}{T} \int_T x(t) e^{-jk\omega_0 t} dt$              |
| DTFS     | $x[n] = \sum_{k=(N)} a_k e^{jk\omega_0 n}$                                                                                                                                                                                                                                                        | $a_k = \frac{1}{N} \sum_{k=(N)} x[n] e^{-jk\omega_0 n}$           |
|          | $\mathbf{x} = \mathbf{E} \mathbf{a}, \mathbf{E} = \begin{bmatrix} 1 & 1 & \cdots & 1 \\ 1 & e^{j\omega_0} & \cdots & e^{j(N-1)\omega_0} \\ \vdots & \vdots & \ddots & \vdots \\ 1 & e^{j(N-1)\omega_0} & \cdots & e^{j(N-1)^2\omega_0} \end{bmatrix}, \mathbf{E}^{-1} = \frac{1}{N} \mathbf{E}^H$ |                                                                   |
| CTFT     | $x(t) = \frac{1}{2\pi} \int_{-\infty}^{\infty} X(j\omega) e^{j\omega t} d\omega$                                                                                                                                                                                                                  | $X(j\omega) = \int_{-\infty}^{\infty} x(t) e^{-j\omega t} dt$     |
|          | 傅里叶积分定理 $x(t) = \frac{1}{2\pi} \int_{-\infty}^{\infty} [\int_{-\infty}^{\infty} x(t) e^{-j\omega t} dt] e^{j\omega t} d\omega$                                                                                                                                                                    |                                                                   |
| DTFT     | $x[n] = \frac{1}{2\pi} \int_{2\pi} X(e^{j\omega}) e^{j\omega n} d\omega$                                                                                                                                                                                                                          | $X(e^{j\omega}) = \sum_{n=-\infty}^{+\infty} x[n] e^{-j\omega n}$ |
| 卷积       | $x[n] * h[n] = \sum_{k=-\infty}^{+\infty} x[k] h[n - k]$                                                                                                                                                                                                                                          | $x(t) * h(t) = \int_{-\infty}^{\infty} x(\tau) h(t - \tau) d\tau$ |
|          | 交换、结合、分配律                                                                                                                                                                                                                                                                                         | 对任一积数时移、微分、积分<br>e.g., $x' * h = h' * x = y'$                     |
| 三角       | $\cos \theta = \frac{1}{2}(e^{j\theta} + e^{-j\theta})$                                                                                                                                                                                                                                           | $\sin \theta = \frac{1}{2j}(e^{j\theta} - e^{-j\theta})$          |
| 内积       | $\langle x(t), y(t) \rangle = \frac{1}{T} \int_T x(t) \cdot y^*(t) dt$                                                                                                                                                                                                                            |                                                                   |
| 奇偶       | $x_{even}(t) = \frac{1}{2}[x(t) + x(-t)]$                                                                                                                                                                                                                                                         | $x_{odd}(t) = \frac{1}{2}[x(t) - x(-t)]$                          |
| 常用积分     | $\int x e^{Ax} dx = \frac{e^{Ax}}{A^2} (Ax - 1)$<br>$\int x^2 e^{Ax} dx = \frac{e^{Ax}}{A^3} (A^2 x^2 - 2Ax + 2)$                                                                                                                                                                                 |                                                                   |
| 微/积分器    | $\delta(t) = u_0(t); u(t) = u_{-1}(t); u_{-n}(t) = \frac{t^{n-1}}{(n-1)!} u(t)$                                                                                                                                                                                                                   |                                                                   |
| 方程       | 两边分别进行傅里叶变换（CT 用微分性质，DT 用时移性质）                                                                                                                                                                                                                                                                    |                                                                   |
| u        | $\delta(t) = u_0(t); u(t) = u_{-1}(t); u_{-n}(t) = \frac{t^{n-1}}{(n-1)!} u(t)$                                                                                                                                                                                                                   |                                                                   |
| δ        | $\delta(at) = \frac{1}{ a } \delta(t); \delta(at - t_0) = \delta\left(a\left(t - \frac{t_0}{a}\right)\right) = \frac{1}{ a } \delta\left(t - \frac{t_0}{a}\right)$<br>$\int_{-\infty}^{\infty} x(t) \delta(at - t_0) dt = \frac{1}{ a } x\left(\frac{t_0}{a}\right)$                              |                                                                   |